AI BOT ASSIGSTANT

CODE

```
"""
AI Assistant - Prompt Engineering Project (FIXED VERSION)
Uses Google's Gemini API
Functions:
1. Answering Questions
2. Summarizing Text
3. Generating Creative Content
"""

import google.generativeai as genai
import os
from datetime import datetime

class AIAssistant:
    def __init__(self):

        self.api_key = os.environ.get("GOOGLE_API_KEY",
"AIzaSyCNwnjU5zPLsAyphMmbl50DNgPgg2sxM0c")
        genai.configure(api_key=self.api_key)

        # Correct and supported model name1
        self.model = genai.GenerativeModel("models/gemini-2.0-flash")
```

```
        self.feedback_log = []

    def get_ai_response(self, prompt, system_message="You are a helpful AI assistant."):
        """Get response from Google Gemini API"""
        try:
            full_prompt = f"{system_message}\n\n{prompt}"
            response = self.model.generate_content(full_prompt)
            return response.text.strip()
        except Exception as e:
            return f"Error: {str(e)}\nPlease ensure your Google API key is set correctly."

    def answer_questions(self):
        """Function 1: Answer factual questions"""
        print("\n=== Question Answering Mode ===")
        question = input("\nEnter your question: ").strip()

        if not question:
            print("No question provided.")
            return

        prompt = f"Provide a clear and factual answer to: {question}"
        system_msg = "You are a knowledgeable assistant that provides accurate, factual
information."
        response = self.get_ai_response(prompt, system_msg)

        print(f"\nAnswer: {response}")
        self.collect_feedback("Question Answering", question, response)
```

```
def summarize_text(self):
    """Function 2: Summarize text"""
    print("\n=== Text Summarization Mode ===")
    print("Enter the text you want to summarize (press Enter twice when done):")

    lines = []
    while True:
        line = input()
        if line == "":
            if len(lines) > 0 and lines[-1] == "":
                break
            lines.append(line)
        else:
            lines.append(line)

    text = "\n".join(lines[:-1])

    if not text.strip():
        print("No text provided.")
        return

    prompt = f"Summarize the following text in 3–5 sentences:\n\n{text}"
    system_msg = "You are an expert at summarizing text concisely while preserving main ideas."
    response = self.get_ai_response(prompt, system_msg)

    print(f"\nSummary: {response}")
    self.collect_feedback("Text Summarization", text[:100] + "...", response)
```

```
def generate_creative_content(self):
    """Function 3: Generate creative content"""
    print("\n=== Creative Content Generation Mode ===")
    prompt_input = input("\nWhat should I create? ").strip()

    if not prompt_input:
        print("No request provided.")
        return

    prompt = f"Create engaging and creative content about: {prompt_input}"
    system_msg = "You are a creative writer with excellent storytelling abilities."
    response = self.get_ai_response(prompt, system_msg)

    print(f"\nCreative Output:\n{response}")
    self.collect_feedback("Creative Content", prompt_input, response)

def collect_feedback(self, function_name, user_input, ai_response):
    """Collect user feedback"""
    print("\n" + "=" * 50)
    feedback = input("Was this response helpful? (yes/no): ").strip().lower()

    feedback_entry = {
        "timestamp": datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
        "function": function_name,
        "user_input": user_input[:100],
        "helpful": feedback == "yes"
    }
```

```
        self.feedback_log.append(feedback_entry)

        if feedback == "yes":
            print("Thank you for your positive feedback!")
        else:
            print("Thank you for your feedback. We'll work on improving!")

    def show_feedback_summary(self):
        """Display feedback summary"""
        if not self.feedback_log:
            print("\nNo feedback collected yet.")
            return

        print("\n=== Feedback Summary ===")
        total = len(self.feedback_log)
        helpful = sum(1 for f in self.feedback_log if f["helpful"])

        print(f"Total interactions: {total}")
        print(f"Helpful responses: {helpful}")
        print(f"Success rate: {(helpful / total) * 100:.1f}%")

        print("\nRecent feedback:")
        for entry in self.feedback_log[-5:]:
            status = "✓" if entry["helpful"] else "✗"
            print(f"{status} [{entry['timestamp']}] {entry['function']}")

    def run(self):
```

```
"""Main loop for the AI Assistant"""
print("=" * 60)
print("     Welcome to Your AI Assistant (FREE)     ")
print("=" * 60)
print("\nThis assistant uses Google Gemini API (FREE)")
print("\nFunctions:")
print("1. Answer factual questions")
print("2. Summarize text")
print("3. Generate creative content")
print("\nNote: Make sure to set your GOOGLE_API_KEY")

while True:
    print("\n" + "=" * 60)
    print("Main Menu")
    print("=" * 60)
    print("1. Answer Questions")
    print("2. Summarize Text")
    print("3. Generate Creative Content")
    print("4. View Feedback Summary")
    print("5. Exit")

    choice = input("\nSelect an option (1-5): ").strip()

    if choice == "1":
        self.answer_questions()
    elif choice == "2":
        self.summarize_text()
    elif choice == "3":
```

```
            self.generate_creative_content()
        elif choice == "4":
            self.show_feedback_summary()
        elif choice == "5":
            print("\nThank you for using the AI Assistant!")
            print("Goodbye!")
            break
        else:
            print("\nInvalid option. Please select 1–5.")

if __name__ == "__main__":
    assistant = AIAssistant()
    assistant.run()
```

USER GUIDE

How to Run the AI Assistant - Quick Start Guide

For New Users

Created by: Bade Naveen Kumar

Project: AI Assistant - Prompt Engineering

🚀 **Quick Start (5 Steps)**

Follow these simple steps to run the AI Assistant on your computer:

Step 1: Install Python

Windows Users:

1. Go to - https://python.org/

2. Click "Downloads"

3. Download Python 3.7+ (or latest version)

4. Run the installer

5. ✅ IMPORTANT: Check the box "Add Python to PATH"

6. Click "Install Now"

7. Wait for installation to complete

Verify Installation:

- Open Command Prompt (search "cmd" in Start Menu)

- Type: python --version

- You should see: Python 3.x.x

Mac Users:

1. Go to https://python.org

2. Download Python 3.7+

3. Run the installer

4. Follow installation prompts

Verify Installation:

- Open **Terminal** (search "Terminal" in Spotlight)

- Type: `python3 --version`

- You should see: `Python 3.x.x`

Linux Users:

Open Terminal and run:

sudo apt update

sudo apt install python3 python3-pip

Step 2: Get Your FREE Google API Key

Follow These Steps:

1. Open your web browser

2. Go to: https://makersuite.google.com/app/apikey

3. Sign in with your Google account (Gmail)

4. Click the blue button: "Create API Key"

5. A popup will appear - Click: "Create API key in new project"

6. Copy your API key - It looks like: AIzaSyAbc123...

7. Save it somewhere safe - You'll need it in Step 4

Important Notes:

- ✅ The API key is **100% FREE**

- ✅ No credit card needed

- ✅ No payment required

- ✅ Keep your API key private (don't share publicly)

Step 3: Install Required Library

Windows:

1. Open Command Prompt (search "cmd")

2. Copy and paste this command:

```
pip install google-generativeai
```

3. Press Enter

4. Wait for installation (10-30 seconds)

5. You should see: Successfully installed google-generativeai

Mac/Linux:

1. Open Terminal

2. Copy and paste this command:

```
pip3 install google-generativeai
```

3. Press Enter

4. Wait for installation

If you see an error, try:

```
python -m pip install google-generativeai
```

Step 4: Setup the AI Assistant Code

Download the Code:

1. Get the file: ai_assistant.py (from your teacher/project folder)

2. Save it to your Desktop or Documents folder

3. Remember the location where you saved it

Add Your API Key:

1. Right-click on ai_assistant.py

2. Open with → Notepad (Windows) or TextEdit (Mac)

3. Find Line 18 - It looks like:

   self.api_key = os.environ.get("GOOGLE_API_KEY", "your-api-key-here")

4. Replace "your-api-key-here"` with your actual API key from Step 2

5. Example:

   self.api_key = os.environ.get("GOOGLE_API_KEY", "AIzaSyAbc123XYZ456...")

   ```

6. Save the file (Ctrl+S on Windows, Cmd+S on Mac)

7. Close the editor

## Step 5: Run the AI Assistant

Windows:

1. Open Command Prompt

2. Navigate to where you saved the file:

   cd Desktop

   (If you saved it in Documents, use: `cd Documents`)

3. Run the program:

   python ai_assistant.py

4. Press Enter

## Mac/Linux:

1. Open Terminal

2. Navigate to the file location:

   cd Desktop

3. Run the program:

   python3 ai_assistant.py

## What You Should See:

==========
   ```

Welcome to Your AI Assistant (FREE)

==========

This assistant uses Google Gemini API (FREE)

Functions:

1. Answer factual questions

2. Summarize text

3. Generate creative content

==========

Main Menu

==========

1. Answer Questions

2. Summarize Text

3. Generate Creative Content

4. View Feedback Summary

5. Exit

Select an option (1-5):

🎉 Congratulations! Your AI Assistant is now running!

📖 How to Use the AI Assistant

Menu Options:

Option 1: Answer Questions

- Type `1` and press Enter

- Ask any question (e.g., "What is AI?")

- Get an instant answer!

- Rate the answer (yes/no)

Option 2: Summarize Text

- Type `2` and press Enter
- Paste or type long text
- Press Enter TWICE when done
- Get a short summary!

Option 3: Generate Creative Content

- Type `3` and press Enter
- Describe what you want (e.g., "Write a story about a robot")
- Get creative content!

Option 4: View Feedback

- Type `4` to see your statistics
- Shows success rate and history

Option 5: Exit

- Type `5` to close the program safely

🆘 Common Problems and Solutions

Problem 1: "python is not recognized"

Solution:

- Python is not installed correctly
- Go back to Step 1 and reinstall
- ✅ Make sure to check "Add Python to PATH" during installation

Problem 2: "No module named 'google.generativeai'"

Solution:

- Run this command again:

  pip install google-generativeai

- Or try:

  python -m pip install google-generativeai

Problem 3: "Invalid API key" or "Authentication failed"

Solution:

- Check your API key in line 18 of the code
- Make sure it starts with `AIza`
- Make sure there are no extra spaces
- Get a new API key from: https://makersuite.google.com/app/apikey

Problem 4: "Error 404 model not found"

Solution:

- Check line 21 in the code
- It should say: `"models/gemini-2.0-flash"`
- Update the library: `pip install --upgrade google-generativeai`

Problem 5: Program is slow

Solution:

- This is normal! AI takes 3-10 seconds to respond
- Check your internet connection
- Wait patiently for the response

Problem 6: Can't find the file

Solution:

- Make sure `ai_assistant.py` is in the correct folder

- Use `cd` command to navigate to the right folder

- On Windows: `cd Desktop` or `cd Documents`

- On Mac: `cd Desktop` or `cd Documents`

⚡ Quick Commands Reference

To Run the Program:

Windows:

```
cd Desktop

python ai_assistant.py
```

Mac/Linux:

```
cd Desktop

python3 ai_assistant.py
```

To Stop the Program:

- Select option `5` from menu (recommended)

- Or press: Ctrl+C (Windows/Linux) or Cmd+C (Mac)

To Update the Library:

```
pip install --upgrade google-generativeai
```

📝 Example Session

Here's what a typical session looks like:

```
Select an option (1-5): 1
```

=== Question Answering Mode ===

Enter your question: What is artificial intelligence?

Answer: Artificial intelligence (AI) is the simulation of human intelligence processes by machines, especially computer systems. These processes include learning, reasoning, and self-correction. AI applications include expert systems, natural language processing, speech recognition and machine vision.

==========

Was this response helpful? (yes/no): yes

Thank you for your positive feedback!

==========

Main Menu

==========

1. Answer Questions
2. Summarize Text
3. Generate Creative Content
4. View Feedback Summary
5. Exit

Select an option (1-5):

✅ Checklist for First-Time Users

Before running, make sure you have:

- Python 3.7+ installed
- Command Prompt/Terminal open
- Google API key obtained
- google-generativeai library installed
- ai_assistant.py file downloaded
- API key added to line 18 of code
- File saved in known location (Desktop/Documents)
- Navigated to correct folder in Command Prompt/Terminal

🎯 Tips for Best Experience

For Questions:

- ✅ Ask clear, specific questions
- ✅ One question at a time
- ✅ Use proper grammar
- ❌ Don't ask multiple questions together

For Summaries:

- ✅ Use text between 50-2000 words
- ✅ Press Enter **TWICE** when done pasting
- ✅ Use well-written text
- ❌ Don't use extremely short text (under 20 words)

For Creative Content:

- ✅ Be specific about what you want
- ✅ Mention style/tone if needed
- ✅ Give clear descriptions

- ❌ Don't be too vague

General Tips:

- ✅ Good internet connection = faster responses
- ✅ Wait 3-10 seconds for AI responses
- ✅ Provide feedback (yes/no) after each response

🌐 Additional Resources

If You Need More Help:

Google AI Studio: https://makersuite.google.com

Python Downloads: https://python.org

Google Gemini Documentation: https://ai.google.dev/docs

Contact:

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Project:AI Assistant - Prompt Engineering

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Institution: SOA University

📄 File You Need

Make sure you have this file:

- ai_assistant.py` - The main program file

File size: ~5 KB

File type: Python script (.py)

Required: Yes (this is the AI Assistant program)

Safe Usage:

- This is a learning project

- Don't use for harmful purposes

- Respect Google's terms of service

- Verify important information from AI responses